

torch.fft.ihfft#

<https://pytorch.org/docs/stable/generated/torch.fft.ihfft.html>

Description:

Computes the inverse of `hfft()`. input must be a real-valued signal, interpreted in the Fourier domain. The IFFT of a real signal is Hermitian-symmetric, $X[i] = \text{conj}(X[-i])$. `ihfft()` represents this in the one-sided form where only the positive frequencies below the Nyquist frequency are included. To compute the full output, use `ifft()`. Supports `torch.half` on CUDA with GPU Architecture SM53 or greater. However it only supports powers of 2 signal length in every transformed dimension. input (Tensor) – the real input tensor

Function Signature

```
torch.fft.ihfft(input, n=None, dim=-1, norm=None, *,  
out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> t = torch.arange(5)
>>> t
tensor([0, 1, 2, 3, 4])
>>> torch.fft.ihfft(t)
tensor([ 2.0000-0.0000j, -0.5000-0.6882j, -0.5000-0.1625j])
```

```
>>> torch.fft.ifft(t)
tensor([ 2.0000-0.0000j, -0.5000-0.6882j, -0.5000-0.1625j,
        -0.5000+0.1625j,
         -0.5000+0.6882j])
```

Notes & Warnings

NOTE

Note Supports torch.half on CUDA with GPU Architecture SM53 or greater. However it only supports powers of 2 signal length in every transformed dimension.

Detailed Documentation

Documentation

Computes the inverse of `hfft()`.

input must be a real-valued signal, interpreted in the Fourier domain. The IFFT of a real signal is Hermitian-symmetric, $X[i] = \text{conj}(X[-i])$. `ihfft()` represents this in the one-sided form where only the positive frequencies below the Nyquist frequency are included. To compute the full output, use `ifft()`.

Supports torch.half on CUDA with GPU Architecture SM53 or greater. However it only supports powers of 2 signal length in every transformed dimension.

input (Tensor) – the real input tensor

n (int, optional) – Signal length. If given, the input will either be zero-padded or trimmed to this length before computing the Hermitian IFFT.

dim (int, optional) – The dimension along which to take the one dimensional Hermitian IFFT.

norm (str, optional) –

Normalization mode. For the backward transform (ihfft()), these correspond to:

"forward" - no normalization

"backward" - normalize by $1/n$

"ortho" - normalize by $1/\sqrt{n}$ (making the IFFT orthonormal)

Calling the forward transform (hfft()) with the same normalization mode will apply an overall normalization of $1/n$ between the two transforms. This is required to make ihfft() the exact inverse.

Default is "backward" (normalize by $1/n$).

out (Tensor, optional) – the output tensor.

Compare against the full output from ifft():